5 Experiments

We evaluate Eso-LMs on two standard language modeling benchmarks: the One Billion Words dataset (LM1B) [3] and OpenWebText (OWT) [6]. For LM1B, we use the `bert-base-uncased` tokenizer [5] and a context length of 128, reporting perplexity on the test split. For OWT, which lacks a standard split, we follow [26, 27, 1] and reserve the last 100K documents for validation, using the `GPT2` tokenizer [24] and a context length of 1024. All models use the transformer architecture from Lou et al. [15], which builds on the diffusion transformer [22] with rotary embeddings [33]. Following Arriola et al. [1], Austin et al. [2], and unlike Sahoo et al. [26], Lou et al. [15], He et al. [7], we apply sequence packing to LM1B, making our setup more challenging and resulting in higher perplexities given the same model (Table 1). Models are trained for 1M steps on LM1B and 250K steps on OWT, without timestep conditioning. Models are trained on H200 GPUs and benchmarked on A6000 GPUs. Full hyperparameters and training details are provided in Suppl. B.6.

5.1 Likelihood Evaluation

Our experiments show that **Eso-LMs achieve state-of-the-art perplexities among diffusion models** on LM1B and OWT, while providing **finer-grained interpolation between MDM and AR perplexities** (Table 1 and Table 2).

Experimental Setup We compare Eso-LMs against leading masked diffusion models—MDLM [26], SEDD [15], D3PM [2], and DiffusionBERT [7]—as well as uniform state models DUO [27], UDLM [29], and specifically BD3-LMs [1], which also interpolate between MDM and AR and support KV caching. All models are trained with `batch_size=512`, consistent with prior work. We split each batch evenly: half trained with the AR loss and half with the MDM loss (7). Refer to Table 5 for an ablation on the split proportion. Refer to Algo. 1 for the training procedure. Attention biases are configured as described in Sec. 4. When training Eso-LMs as a pure MDM ($\alpha_0 = 1$), the full batch is trained with the MDM loss. For this setting only, we replace the diffusion coefficient $\alpha'_t/(1-\alpha_t)$ with -1, which empirically reduced training variance and improved convergence.

Results For our method, PPL is computed using (7). On LM1B, we train Eso-LM (A) with $\alpha_0 \in \{0.0625, 0.125, 0.25, 0.5, 1.0\}$, and Eso-LM (B) with $\alpha_0 = 1$. We find that Eso-LM (A) effectively interpolates between MDLM and AR perplexities. Surprisingly, Eso-LM (A), which modifies MDMs by replacing bidirectional attention over mask tokens with a causal mask, outperforms MDLM by ≈ 1 PPL at $\alpha_0 = 1$. This improvement can be attributed to the low-variance training loss as described earlier. In contrast, Eso-LM (B) introduces causal dependencies among clean tokens, which degrades performance by ≈ 4 PPL compared to Eso-LM (A) at $\alpha_0 = 1$. This is expected because bidirectional attention on the clean tokens in Eso-LM (A) is strictly more expressive than causal attention in Eso-LM (B). On OWT, Eso-LM (B) is particularly compelling as it is the only model—BD3-LMs included—that supports KV caching during the diffusion phase. Trained with $\alpha_0 \in \{0.0625, 0.125, 0.25, 0.5, 1.0\}$, it shows smooth interpolation between AR and MDM perplexities (Table 2). As with LM1B, Eso-LM (A) remains competitive with MDLM on OWT.

5.2 Generation Speed

Our qualitative evaluation experiments show that **Eso-LMs are 3 – 4× faster than prior diffusion based methods that support KV caching** and about **up to 65× faster than MDMs that don't support KV caching**.

Experimental Setup Diffusion models like MDLM do not support KV caching due to their use of bidirectional attention. Semi-autoregressive models such as BD3-LMs enable partial KV caching by generating blocks of tokens autoregressively and caching completed blocks, but they still lack caching for the current block under diffusion. In contrast, our method uniquely supports KV caching during the diffusion phase. We compare inference times of our method, Eso-LM (B), against MDLM and BD3-LMs with context lengths $L \in \{2048, 8192\}$, using the first-hitting sampler [37], and a batch size of 1. To simulate the worst-case scenario, we set $T \gg L$ to ensure all methods have approximately L NFEs: $T = 1$M for MDLM and Eso-LM (B), $T' = 5000$ (number of sampling steps per block) for BD3-LMs. We found that nucleus sampling yields a non-negligible overhead for all methods, and hence disable it to focus on the relation between sampling speed and sequence length.

Table 1: Test perplexities (PPL; ↓) on LM1B. *Reported in He et al. [7]. ¶Denotes the dataset didn't incorporate sentence packing. †Reported in Arriola et al. [1]. For diffusion models, we report the bound on the likelihood. ‡Reported in Sahoo et al. [27].

	PPL (↓)
Autoregressive	
Transformer‡	22.83
Diffusion	
D3PM Uniform [2]	137.90¶
D3PM Absorb [2]	76.90¶
Diffusion-LM* [12]	118.62¶
DiffusionBert [7]	63.78
SEDD Absorb‡ [15]	32.71¶
SEDD Uniform¶ [15]	40.25¶
MDLM‡ [26]	31.78
UDLM‡ [29]	36.71
DUO‡ [27]	33.68
Interpolating diffusion and autoregressive	
BD3-LMs† [1]	
$L' = 16$	30.60
$L' = 8$	29.83
$L' = 4$	28.23
Eso-LM (B) (Ours)	
$\alpha_0 = 1.0$	35.00
Eso-LM (A) (Ours)	
$\alpha_0 = 1.0$	30.96
$\alpha_0 = 0.5$	30.51
$\alpha_0 = 0.25$	28.44
$\alpha_0 = 0.125$	25.97
$\alpha_0 = 0.0625$	**24.51**

Table 2: Test perplexities (PPL; ↓) on OWT for models trained for 250K. ‡ Reported in [1]. † Denotes retrained models. ¶ Intermediate checkpoints were provided by Sahoo et al. [26].

	PPL (↓)
Autoregressive	
Transformer	17.90¶
Diffusion	
SEDD Absorb [15]	26.81¶
MDLM [26]	25.76¶
UDLM [29]	30.52¶
DUO [27]	27.14¶
Interpolating diffusion and autoregressive	
BD3-LMs [1]	
$L' = 16$	23.57†
Eso-LM (A) (Ours)	
$\alpha_0 = 1$	26.21
Eso-LM (B) (Ours)	
$\alpha_0 = 1$	30.14
$\alpha_0 = 0.5$	27.85
$\alpha_0 = 0.25$	24.73
$\alpha_0 = 0.125$	**21.87**

Table 3: Sampling time (↓) in seconds for sequence lengths $L \in \{2048, 8192\}$ with NFEs set to L for all methods. Reported values are mean_{std} over 5 runs.

Method	$L = 2048$	$L = 8192$
AR (Oracle)	$13.3_{0.9}$	$54.0_{0.2}$
MDLM	$201.3_{0.4}$	$5438.3_{3.3}$
BD3-LM ($L' = 4$)	$24.3_{0.7}$	$312.0_{1.7}$
BD3-LM ($L' = 16$)	$21.3_{0.1}$	$268.1_{1.2}$
Eso-LM (B)	$\mathbf{14.6}_{0.3}$	$\mathbf{82.1}_{0.3}$

Results As shown in Table 3, as compared to MDLM which lacks KV caching, Eso-LM (B) is ~14× faster for $L = 2048$, and ~65× faster for $L = 8192$. Compared to BD3-LMs, which partially support caching, Eso-LM (B) is ~3.2× faster than BD3-LM ($L' = 16$) and ~3.8× faster than BD3-LM ($L' = 4$) at $L = 8192$. This speedup stems from KV-caching and the scheduler $\mathcal{S}$ that restricts the forward pass to the masked tokens that are supposed to be denoised and previously predicted tokens, avoiding redundant computation—a feature BD3-LMs lack. As we restrict the NFEs to L, our method is slower than AR models due to delayed KV reuse—only possible from the penultimate denoising step—its overall efficiency still surpasses prior diffusion-based baselines (see Sec. 4.2.2).

5.3 Generation Quality

Our sampling experiments show that (1) **Eso-LMs establish a new SOTA on the Pareto frontier of sampling speed and quality** (Fig. 4), (2) **generate higher-quality samples than all previous diffusion models at high NFEs**, and (3) **don't suffer mode collapse at low NFEs** unlike the previous interpolating diffusion method BD3-LMs (Fig. 3).

Experimental Setup We train Eso-LM (B) with $\alpha_0^{\text{train}} \in \{0.125, 0.25, 1\}$ and generate samples with varying $(\alpha_0^{\text{eval}}, T) \in \{0.0625, 0.25, 0.5, 1\} \times \{16, 128, 1024\}$ to control NFEs (NFEs = $|\mathcal{S}|$). MDLM and BD3-LMs use ancestral sampling as proposed in Sahoo et al. [26], with $T \in \{16, 128, 256, 512, 1024, 4096\}$ for MDLM and $T \in \{16, 128, 256, 512, 1024, 2048, 4096\}$ for BD3-LMs. All generations are $L = 1024$ tokens long. BD3-LMs are evaluated with block sizes $L' \in \{4, 8, 16\}$ and $T' = T/(1024/L')$. Consequently, $T = 128$ is not applicable to BD3-LMs with $L' = 4$ and $T = 16$ is not applicable to all BD3-LMs considered, since these would result in T' values